

Software Requirements Specification

Project: NBP Data Analysis Tool

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The primary goal of the IT system is to provide users with a reliable, terminal-based tool for analyzing historical currency exchange rates. It aims to process data retrieved from the NBP API to deliver precise statistical measures and session trends, assisting users in tracking currency fluctuations over specific timeframes.

1. Functional requirements

- **API integration:** The system shall connect to the NBP API (<http://api.nbp.pl/>) to fetch current and historical currency exchange rates.
- **User input and parameter selection:** The system shall allow the user to select:
 - type of statistical analysis,
 - specific currency (or pair of currencies),
 - time period for the calculation.
- **Available analysis time periods:**
 - The system shall enable analysis over following intervals:
1 week, 2 weeks, 1 month, 1 quarter, 6 months, 1 year.
- **Types of statistical analysis:**
 - **Session analysis:** The system shall calculate and return the total number of rising sessions, losing sessions, and flat sessions for the selected currency and time period.
 - **Statistical measures:** The system shall calculate and return the following metrics: median, mode, standard deviation, coefficient of variation for the selected currency and time period.
 - **Change distribution:** The system shall calculate the distribution of changes for user-selected currency pairs and time period (only quarterly or monthly). It shall calculate session-to-session changes and sum them up over the selected period. The return value shall be presented as a frequency histogram displaying the number of changes within specific value ranges.
 - **Supported currencies:** The system shall support the analysis of the currencies specified in the table A of the API (<https://nbp.pl/statystyka-i-sprawozdawczosc/kursy/tabela-a/>). For change distribution, supported pairs include combinations of these currencies (e.g., EUR/USD), also including PLN (which is not listed in the table A).
 - **Data presentation and export:** The system shall present the results of the analysis as a table in the terminal. Additionally, it shall provide an option to export the generated results to a CSV file.

2. Non-functional requirements

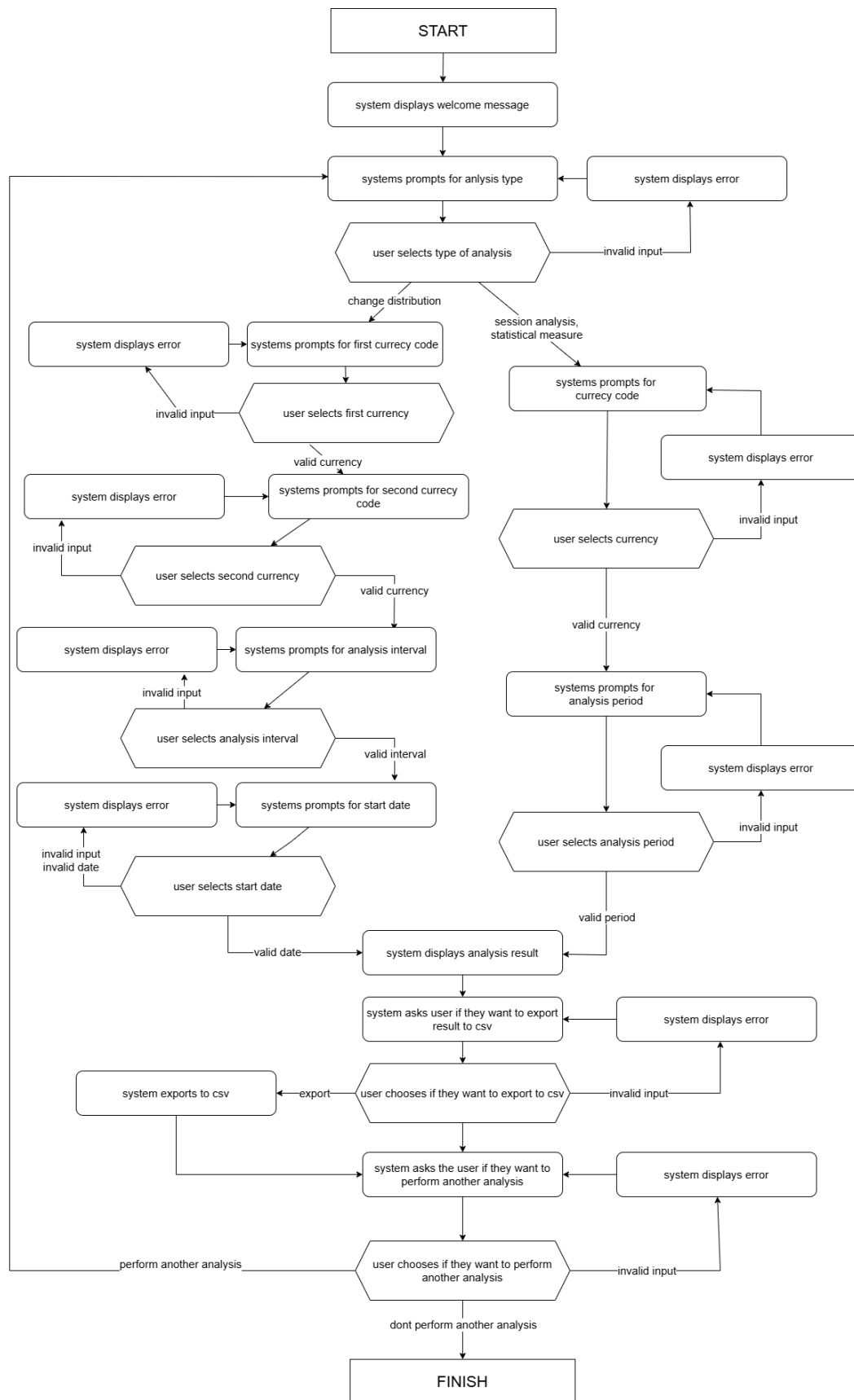
- **Security (protection against unauthorized access):**
 - As a locally run application, it does not require user authentication or a login module.
 - All external communication with the NBP API shall be secured using the HTTPS protocol to ensure data integrity and protect against man-in-the-middle attacks.
- **Usability (ease of use):**
 - The command-line interface (CLI) shall be intuitive, providing clear textual prompts.
 - The system shall validate user inputs and print clear, non-technical error messages directly in the terminal in case of incorrect parameters, typos, or lack of network connection.
- **CLI communication flow:**
 - The user-program communication shall operate via a sequential, interactive terminal prompt.
 - The communication shall be kept entirely in textual format. For better readability output shall be color coded:
prompt, analysis results - white
available input options – yellow
correct input acknowledgement – green
error message – red
 - After the system prompts the user for input, available input options shall be displayed in the console.
Example:
1- Session analysis
2- Statistical measures
3- Change distribution
 - The user shall be informed about any incorrect input by an error message with explanation.
Example:
INPUT INVALID (please enter a number)
 - In case of any incorrect user input, the system shall repeat the current step, until the user enters a valid choice.
 - The resulting analysis data shall be presented in the form of an ascii table.
Sample ascii table:

A	B	C	D
1	2	3	4
1	2	3	4

○ Selection options specification

Selection	Available options	Possible error messages
Analysis type	1- Session analysis 2- Statistical measure 3- Change distribution	INPUT INVALID (only one character allowed) INPUT INVALID (please enter a number) INPUT INVALID (number not in available types)
currency	THB, USD, AUD, HKD, CAD, NZD, SGD, EUR, HUF, CHF, GBP, UAH, JPY, CZK, DKK, ISK, NOK, SEK, RON, TRY, ILS, CLP, PHP, MXN, ZAR, BRL, MYR, IDR, INR, KRW, CNY, XDR, PLN	INPUT INVALID (currency code not recognized)
first currency		INPUT INVALID (currency code not recognized) INPUT INVALID (second currency cannot be the same as first)
second currency		
Analysis period	1 - one week 2 - two weeks 3 - one month 4 - one quarter 5 - six months 6 - one year	INPUT INVALID (only one character allowed) INPUT INVALID (please enter a number) INPUT INVALID (number not in available periods)
Analysis interval	1- monthly 2- quarterly	INPUT INVALID (enter 1 or 2)
Analysis start date	date (format: DD.MM.YYYY)	INPUT INVALID (wrong date format) INPUT INVALID (date insufficient, earliest allowed: <i>[today - interval]</i>) INPUT INVALID (date cannot be in the future)
Export to csv?	Y N	INPUT INVALID (enter y or n)
Perform another analysis?		

Communication flow diagram

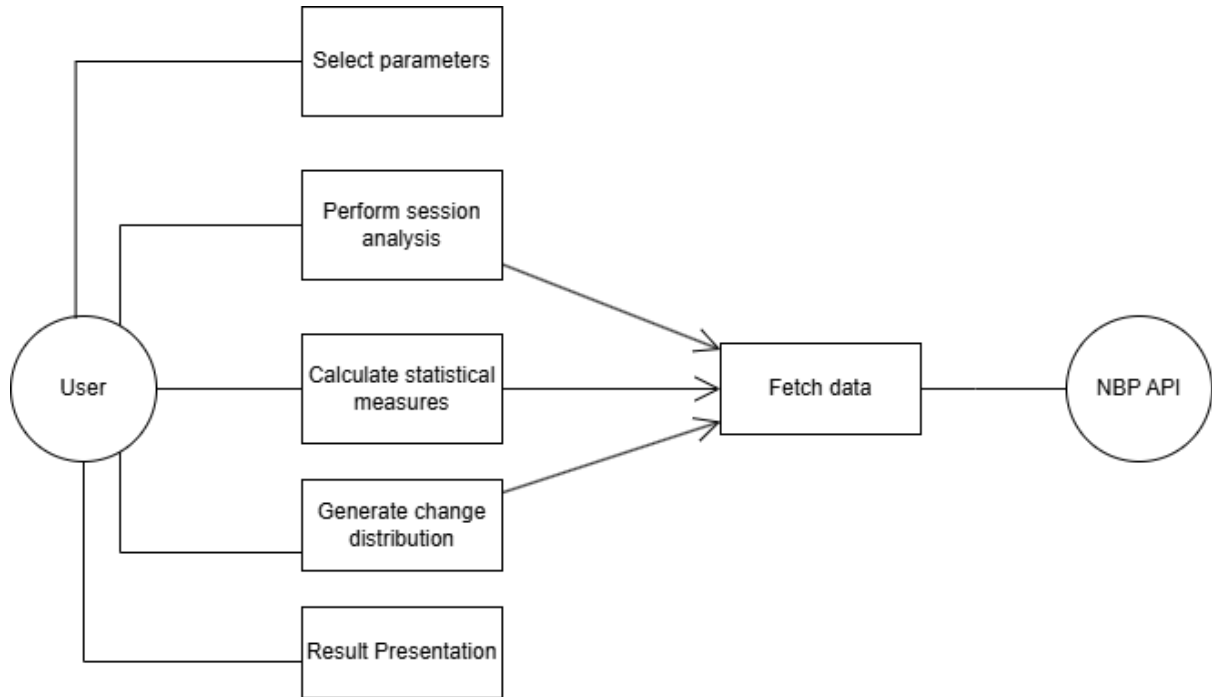


- **Reliability:**
 - The system shall not crash due to external errors. It shall gracefully handle NBP API connection timeouts, missing data and invalid API responses without terminating the process abruptly.
 - Test coverage (unit and functional tests) shall be maintained at a high level. Furthermore, the system shall implement integrity tests to ensure data consistency throughout the processing pipeline.
 - NBP API Limitations: The system logic shall account for external functionality limitations of the NBP API:
 - Data is not published on weekends and public holidays; the system shall handle these gaps mathematically without throwing errors.
 - The system shall handle potential API rate limits gracefully.
 - The API is only accessible via HTTPS.
 - The system shall gracefully handle external API unavailability without causing failures in core functionality.
- **Portability (ability to run in different environments):**
 - The system's source code shall be cross-platform, capable of running via a standard Python interpreter on Windows, macOS, and Linux. However, the automated CI/CD deployment shall exclusively provide a pre-compiled standalone executable (.exe) for Windows users attached to the GitHub Release (see point "Deployment and installation").
- **Language:**
 - All source code, including variable names, functions, and inline comments, shall be written strictly in English.
 - All project documentation, including the readme.md, commit messages, pull requests, bug reports, and issue tracking in the product backlog, shall be maintained entirely in English.
 - The command-line interface (CLI) outputs, including prompts, results, and error messages, shall be presented in English.
- **Tech stack:**
 - **Programming language:** Python 3.x (console application).
 - **Data processing:** Standard Python libraries, external Python libraries.
- **Deployment and installation**
 - The application shall be developed as a Python command-line tool.
 - As part of the CI/CD pipeline, every approved pull request merged into the main branch will automatically trigger a build process.

- The CI/CD pipeline shall utilize PyInstaller to compile the Python source code into a standalone executable file (.exe for Windows environments).
- This standalone executable shall be automatically attached as a deployment artifact to the newly generated and tagged GitHub Release, allowing users to run the application natively without installing Python or any external dependencies.
- **Warranty & maintenance**

The product is out of warranty, and no maintenance or further support shall be provided.

3. Use case diagram



4. Gantt chart

